Free groups as words: the set-quotient of lists of letters and formal inverses is the free group

The Univalent Foundations Program

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Editorial note. The counted theorem carries no \label in the pinned source; its identity is fixed by the pinned file plus the original number 6.11.7, obtained from the label-number dump and by counting the theorem-class environments of section 6.11 of the same file (the neighbouring labelled results match: thm:free-monoid is 6.11.5 and thm:freegroup-nonset is 6.11.8).

Editorial note. The proof environment begins on the line immediately after \end{thm}, so the pairing is by adjacency, and the retained proof range 1648-1678 is the whole environment.

Editorial note. The retained context is the author's own lead-in paragraph introducing words, including the formal-inverse notation that the statement uses.

Editorial note. The retained counter values (chapter 6, section 11, theorem offset 6) make the retained theorem print as Theorem 6.11.7, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: the free group presented by words, Theorem 6.11.7.

Allowed standard prerequisites (editorial).

- Set-quotients by a generated equivalence relation. A set-quotient may be presented by relations, and a map out of it is defined by quotient recursion once the map respects the generating relations.. The theorem defines F'(A) as such a quotient at source line 1639, and the proof defines multiplication and inversion on it by quotient recursion at lines 1651 and 1659.
- Lists over a type form a monoid. lst B with concatenation and the empty list is a monoid, and it has a recursion principle by list induction.. The monoid structure of lst(A+A) is what descends to the quotient at source line 1650, and list induction is used at lines 1662-1669.
- Quotient induction and double quotient recursion. To prove a family of propositions over a set-quotient it suffices to prove it on representatives, and a binary operation may be defined by double quotient recursion when

it respects the relation in each variable.. Double quotient recursion defines the multiplication at source line 1651 and quotient induction reduces the inverse law to lists at line 1662.

Of course, it is sometimes also useful to have a concrete description of free algebraic structures. In the case of free groups, we can provide one, using quotients. Consider $\mathsf{List}(A + A)$, where in $A + A$ we write $\mathsf{inl}(a)$ as a, and $\mathsf{inr}(a)$ as $\hat{a}$ (intended to stand for the formal inverse of a). The elements of $\mathsf{List}(A + A)$ are *words* for the free group on A.

Theorem 6.11.7. *Let A be a set, and let $F'(A)$ be the set-quotient of $\mathsf{List}(A+A)$ by the following relations.*

$$(\dots, a_1, a_2, \widehat{a_2}, a_3, \dots) = (\dots, a_1, a_3, \dots)$$
$$(\dots, a_1, \widehat{a_2}, a_2, a_3, \dots) = (\dots, a_1, a_3, \dots).$$

Then $F'(A)$ is also the free group on the set A.

Proof. First we show that $F'(A)$ is a group. We have seen that $\mathsf{List}(A + A)$ is a monoid; we claim that the monoid structure descends to the quotient. We define $F'(A) \times F'(A) \to F'(A)$ by double quotient recursion; it suffices to check that the equivalence relation generated by the given relations is preserved by concatenation of lists. Similarly, we prove the associativity and unit laws by quotient induction.

In order to define inverses in $F'(A)$, we first define $\mathsf{reverse} : \mathsf{List}(B) \to \mathsf{List}(B)$ by recursion on lists:

$$\mathsf{reverse}(\mathsf{nil}) :\equiv \mathsf{nil},$$
$$\mathsf{reverse}(\mathsf{cons}(b, \ell)) :\equiv \mathsf{reverse}(\ell) \cdot \mathsf{cons}(b, \mathsf{nil}).$$

Now we define $i : F'(A) \to F'(A)$ by quotient recursion, acting on a list $\ell : \mathsf{List}(A+A)$ by switching the two copies of A and reversing the list. This preserves the relations, hence descends to the quotient. And we can prove that $i(x) \cdot x = e$ for $x : F'(A)$ by induction. First, quotient induction allows us to assume x comes from $\ell : \mathsf{List}(A + A)$, and then we can do list induction; if we write $q : \mathsf{List}(A + A) \to F'(A)$ for the quotient map, the cases are

$$\begin{aligned}
i(q(\mathsf{nil})) \cdot q(\mathsf{nil}) &= q(\mathsf{nil}) \cdot q(\mathsf{nil}) \\
&= q(\mathsf{nil}) \\
i(q(\mathsf{cons}(a, \ell))) \cdot q(\mathsf{cons}(a, \ell)) &= i(q(\ell)) \cdot q(\mathsf{cons}(\hat{a}, \mathsf{nil})) \cdot q(\mathsf{cons}(a, \ell)) \\
&= i(q(\ell)) \cdot q(\mathsf{cons}(\hat{a}, \mathsf{cons}(a, \ell))) \\
&= i(q(\ell)) \cdot q(\ell) \\
&= q(\mathsf{nil}). \qquad \text{(by the inductive hypothesis)}
\end{aligned}$$

(We have omitted a number of fairly evident lemmas about the behavior of concatenation of lists, etc.)

This completes the proof that $F'(A)$ is a group. Now if G is any group with a function $f : A \to G$, we can define $A + A \to G$ to be f on the first copy of A and f composed with the inversion map of G on the second copy. Now the fact that G is a monoid yields a monoid homomorphism $\mathsf{List}(A + A) \to G$. And

since G is a group, this map respects the relations, hence descends to a map $F'(A) \to G$. It is straightforward to prove that this is a group homomorphism, and the unique one which restricts to f on A. $\square$